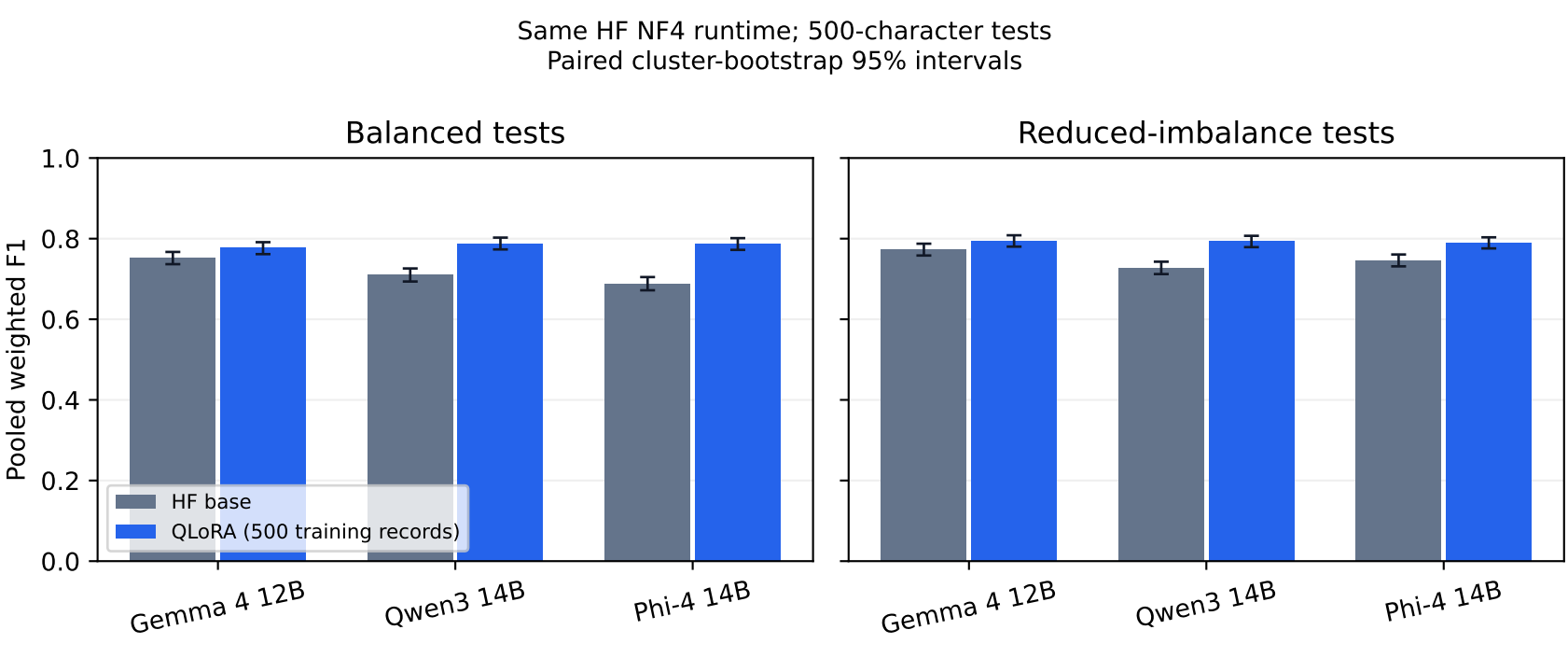
Same HF NF4 runtime; 500-character tests
Paired cluster-bootstrap 95% intervals
Balanced tests
Reduced-imbalance tests
Pooled weighted F1
1.0
0.8
0.6
0.4
0.2
0.0
HF base
QLoRA (500 training records)
Gemma 4 12B
Qwen3 14B
Phi-4 14B
Gemma 4 12B
Qwen3 14B
Phi-4 14B